Module 1: Algebra & Functions

Ph.D. Math Camp

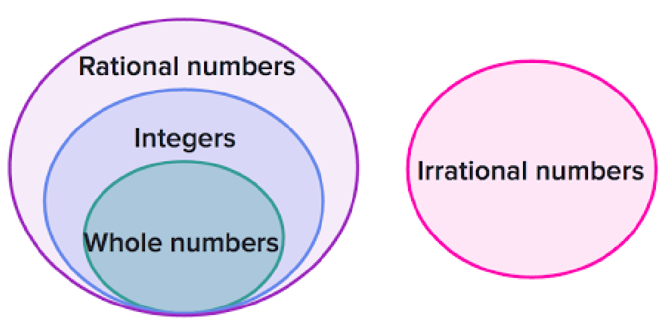
1 Algebra Review

Numbers. We can have **counting numbers** or **natural numbers** ($\mathbb{N}$): 1, 2, 3, 100, 50000, ... or **Whole Numbers** which includes 0 together with the counting numbers.

We also have **Integers** ($\mathbb{Z}$): are whole numbers but can be negative (-1,-2,-3,...).

Rational Numbers ($\mathbb{Q}$): ratio of two integers (i.e., a rational number can be written as a fraction). If you can write a number as a ratio of two integers (e.g., 1 over 10, -5 over 23, 1 over 3, 1543 over 10, etc.), it is a rational number. If we cannot, then we say it is **Irrational Numbers**. These would be decimals that do not stop and do not repeat. Some important irrational numbers are π or e.

Figure 1: Types of Numbers



Lastly, **Real Numbers** ($\mathbb{R}$) are all rational and irrational numbers.

- $\mathbb{R}^+$: set of all positive real numbers

- $\mathbb{R}^-$: set of negative real numbers

1.1 Inequalities and Intervals

- An inequality is a comparison of unequal terms

- Real number a is ____ real number b

  - Greater than $>$
  - Less than $<$
  - Greater than or equal to $\geq$
  - Less than or equal to $\leq$

Intervals are a set of real numbers that can be represented on a number line by a line segment. Inequalities are often used to describe intervals.

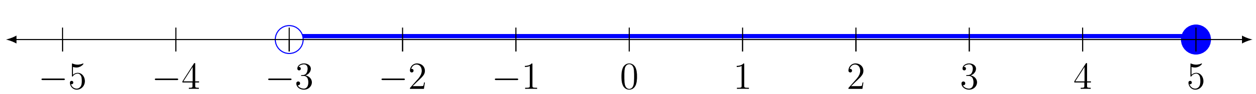
- Intervals may be finite or infinite in extent and may or may not contain either endpoint.

- Often intervals are written in brackets and parentheses

- e.g., Interval from 3 to 5.

  - To show inclusive of endpoints, use brackets: [3,5]
  - Exclusive of the endpoints use parenthesis: (3,5)
  - Inclusive of the right endpoint only use the parenthesis on the left and the bracket on the right (3, 5]

- On a number line, an interval inclusive of the endpoints has solid dots on the endpoints, whereas an interval exclusive of the endpoints has open dots on the endpoints. Take (-3, 5] as an example:



1.2 Absolute Value

The absolute value of a number can be interpreted geometrically as its **distance from zero on a number line.**

- Also, the distance between any two numbers is **the absolute value of their difference** (in either order). To see that $|a - b|$ is the difference between a and b and the distance of $a - b$ from the origin, examine a few pairs of a and b on the number line.
- Absolute value is represented as $|\ \ |$

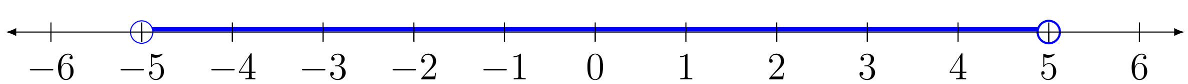
For the interval (-3,5] what is the distance between the endpoints?

$$|-3-5| \ = \ |5+3| = 8$$

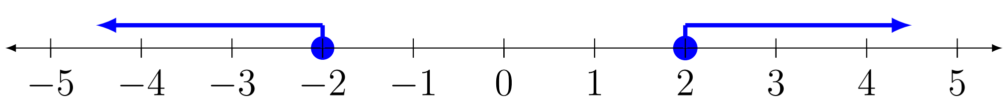
Combining Absolute Value and Inequalities

Let's combine the concepts of absolute value and inequalities and solve for x in some basic expressions involving two operations.

1. Find the set of solutions for $|x| < 5$. In English: **We are looking for real numbers x whose distance from the origin (zero) is less than 5 units.** This is the interval (-5,5).



2. Solutions to $|x| \geq 2$. In English: which numbers, x, are at least 2 units **away** from the origin? We can write this in interval notation $(-\infty, -2] \cup [2,\ \infty)$.



3. Find the solution to $|x - 2| \leq 1$

$$-1 \le x - 2 \le 1.$$
$$\text{Solve for } x\text{: } 2 - 1 \le x \le 1 + 2 \rightarrow \boxed{1 \le x \le 3}$$

4. Find the solution to $|2x - 5| < 2$

$$-2 < 2x - 5 < 2. \text{ Solve: } \quad -2 + 5 < 2x < 2 + 5 \implies \boxed{\frac{3}{2} < x < \frac{7}{2}}$$

1.3 Exponential Notation

The following rules define the expression a^x for $a > 0$ and all rational values of x:

- Integer Powers: If n is a positive integer, then $a^n = a \cdot a \cdot a \cdot \cdots \cdot a$, where this product contains n factors.
- Fractional Powers: If n and m are positive integers, then $a^{\frac{n}{m}} = (\sqrt[m]{a^n}) = (\sqrt[m]{a})^n$
- Negative powers: $a^{-x} = \frac{1}{a^x}$
- Zero power: $a^0 = 1$

Remember these as you will use them repeatedly.

Some examples:

a) $3^0 = 1$

b) $27^{-2/3} = (27^{\frac{1}{3}})^{-2} = (3)^{-2} = \frac{1}{9}$

Laws of Exponents

If a is a real number

- The product law: $a^r \cdot a^s = a^{r+s}$
- The quotient law: $\frac{a^r}{a^s} = a^{r-s}$
- The power law: $(a^r)^s = a^{rs}$

Again, very important to remember these.

Some examples:

a) $\frac{1}{r^5} \cdot \frac{1}{r^5} = r^{-10}$

b) $\beta^7 \cdot \beta^{-3} = \beta^4$

1.4 Factoring

1. Distributive law: For any real numbers a, b, and c: $\quad ab + ac = a(b + c)$.

   The key to factoring is spotting common terms. You have to practice to get good.

2. Polynomials: You will most often be factoring polynomials, expressions of the form $a_0 + a_1x + a_2x^2 + ... + a_nx^n$.

   Here x is a variable, n is a nonnegative integer, and $a_0, a_1, a_2...a_n$ are real numbers or coefficients of the polynomial. If $a_n \neq 0,\ n$ is said to be the **degree** of the polynomial. For example, $8c^7 + 6c^4 + c^2 + 8$ is a polynomial of degree 7.

   - *Variables* and *constants* can be expressed by upper or lower case letters and allow use in equations to help explain concepts within theories.
   - *Constants* do not vary in an expression or across individuals or units of observations. *Variables* can take on different values. For instance, we may have a coefficient on a variable we are trying to solve for that can take on different values. Variables can take on different values within a given set.

3. To factor an expression is to write it as a product of two or more simpler terms, called **factors**. Factoring a natural number means expressing it as a product of smaller natural numbers. Factoring a polynomial means expressing it as a product of polynomials of a lower degree. Factoring is used to simplify complicated expressions and solve equations and is based on the distributive law of multiplication over addition.

We will most likely be looking at polynomials of degree 2 (also known as quadratics).

The goal of factoring a quadratic is to go from an expression of the form $a_2x^2 + a_1x + a_0$ to one of the form $(cx + a)(dx + b)$, with a, b, c, d being integers.

Clearly, c and d are factors of a_2. Also, a and b are factors of a_0. As a short-cut that sometimes works, you want to find two numbers that, when multiplied, equal to $a_2 \times a_0$ and, when added, equal to a_1. The best way to learn is via examples:

a) $x^2 + 11x + 24$. In this case, the two numbers are 8 and 3. Set up the next step of the solution as $(x + 8)(x + 3)$.

b) $p^2 - 22p + 40$. The two numbers are -2 and -20 $\implies$ $(p - 2)(p - 20)$.

There is also a special kind of polynomial that **is the difference of two perfect squares**

- For any real numbers a and b, $a^2 - b^2 = (a + b)(a - b)$. Notice how the ab terms cancel when you multiply the RHS.

Another key way that you can use factoring is in simplifying fractions by factoring both the numerator and the denominator and divide each by common terms.

Example:

$$\frac{7(q+4)^2(q-1)^4 - 14(q+4)^3(q-1)^2}{(q+4)(q-1)^2}$$

$$= 7(q + 4)[(q - 1)^2 - 2(q + 4)]$$

Factor a $(q + 4)$ and $(q - 1)^2$ from top and bottom.

Simplify the following expression $\frac{3(x-2)^2(x+1)^2 - 2(x-2)(x+1)^3}{(x-2)^4}$

1.5 Solving Equations using Factoring

Sometimes we can solve equations using factoring.

Some Examples:

a) $x^2 + x - 6 = 0$ $(x+3)(x-2) = 0 \implies (x+3) = 0$ and/or $(x-2) = 0$

b) $y^2 - 10y + 16 = 0 \implies (y-8) = 0$ and/or $(y-2) = 0$

c) $(s^2 - 81) = 0 \implies (s+9)(s-9) = 0$

Formally, a **quadratic equation** takes the form $ax^2 + bx + c = 0 \quad (a \neq 0)$.[1]
It can be shown that any polynomial of degree n can be factored into n linear (degree 1) expressions, if we allow complex and repeated roots. A quadratic equation, therefore, can be factored into 2 linear expressions. Thus, we say that this type of equation has at most two real solutions or 'roots.'

If we are unable to factor the equation with integer coefficients to find the solution, then we can always use the quadratic formula to solve quadratic equations.
Find the solution or roots with the following formula:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

The term $D = b^2 - 4ac$ is called the **discriminant** of the quadratic equation. The sign of D informs us on the number of solutions and the type of solutions.

- $D > 0$ means the equation has two distinct real solutions
- $D = 0$ means the equation has one distinct real solution (a real double root)– bounces off or just touches x-axis
- $D < 0$ means the equation has two distinct complex conjugate solutions.

[1]The quadratic formula covering all cases was first obtained by Simon Stevin in 1594. In 1637 René Descartes published La Géométrie containing special cases of the quadratic formula in the form we know today.

Example:

a) $3x^2 - 41x + 26 = 0$

$$x = \frac{41 \pm \sqrt{41^2 - 4(3)(26)}}{2(3)} = \frac{41 \pm \sqrt{1681 - 312}}{6}$$

$$x = \frac{41 \pm \sqrt{1369}}{6} = \frac{41 \pm 37}{6}$$

$$\Longrightarrow x = \frac{41+37}{6};\; x = \frac{41-37}{6}$$

$$\Longrightarrow x = \frac{78}{6};\; x = \frac{4}{6}$$

$$\Longrightarrow \boxed{x = 13}\, \boxed{x = \frac{2}{3}}$$

1.6 System of Equations

A group of equations to be solved is called a system of equations. Geometrically, each equation represents a surface in some space. The coordinates of any point on the surface solve that equation. A point solves the system of equations if it is on all surfaces. Therefore, the common intersection of all surfaces contains all the solutions to the system of equations.

Linear equations (those representing straight lines) are much less complicated than non-linear equations. And the simplest linear system is one with two equations and two variables.

We generally don't find solutions to equations by simply substituting random points and checking to see if they work in an equation.

There are two ways to solve systems like these: (1) Elimination by multiplication/addition ; (2) elimination by substitution.

Example: Find real numbers p and q that satisfy the system

$$\begin{aligned} &(1) \quad 8p - 3q = 7 \\ &(2) \ -p + 7q = 19 \end{aligned}$$

Let's start with **Elimination by multiplication/addition**:

Step 1: Let's decide to eliminate p by multiplying equation (2) by 8. Note that by multiplying both sides of the equality we aren't changing the solution.

$$(2a) \implies -8p + 56q = 152$$

Step 2: Now add (1) and (2a) to get $53q = 159$ now solve for $q \implies \boxed{q = 3}$

Step 3: Plug $q = 3$ into either (1) or (2) to get $-p + 7(3) = 19 \implies \quad -p = -2 \implies$ $\boxed{p = 2}$

Now let's solve the same system using the second method: **Elimination by substitution**.

Step 1: Solve either (1) or (2) for either p or q. Let's use (2) to solve for p.

$$-p + 7q = 19 \implies -p = 19 - 7q \implies p = 7q - 19$$

Step 2: Plug the new expression for p into (1) and solve for q.

$$8(7q - 19) - 3q = 7 \implies 56q - 152 - 3q = 7 \implies 53q = 159 \implies \boxed{q = 3}$$

Step 3: Plug $q = 3$ into (2) to get $-p + 7(3) = 19 \implies -p = -2 \implies \boxed{p = 2}$

Notice that we get identical answers regardless of which method we use.

Example: Try another one.

$$\begin{aligned} &(1) \qquad 2x + y = 5 \\ &(2) \ -x + 4y = -7 \end{aligned}$$

1.7 Changing Parameters in Systems of Two Equations and Two Unknowns

Let's consider two lines in the plane represented by the following two linear equations

$$a_1x + b_1y + c_1 = 0 \tag{1}$$

$$a_2x + b_2y + c_2 = 0 \tag{2}$$

The form of this equation of a line is called the **canonical form**. You may be more familiar with the slope/intercept form. It is not difficult to switch between the two and graph them.

For example, equation (1) can be transformed to slope/intercept form by subtracting a_1x and c_1 from both sides of the equation and dividing by b_1. The result is

$$y = (\frac{-a_1}{b_1})x + (\frac{-c_1}{b_1}) \tag{3}$$

For the second equation,

$$y = (\frac{-a_2}{b_2})x + (\frac{-c_2}{b_2}) \tag{4}$$

In equation (3) the slope of the line is $m = \frac{-a_1}{b_1}$ and the y-intercept (the vertical coordinate of the point of intersection of the line and the y-axis) is $\frac{-c_1}{b_1}$

Given two lines, there are three possible situations that can occur:

1. Case 1: The lines are identical: this happens when the equations are identical (when corresponding coefficients are the same multiple of each other): $a_2 = ka_1,\ b_2 = kb_1$; and $c_2 = kc_1$.

   Let's consider

   $$4x + 2y - 8 = 0 \tag{5}$$

   $$6x + 3y - 12 = 0. \tag{6}$$

   Notice that the coefficients of (6) are 1.5 times the coefficients of equation (5).

$$6x + 3y - 12 = 0 \implies \quad 3y = -6x + 12 \implies y = -2x + 4$$

$$4x + 2y - 8 = 0 \implies \quad 2y = -4x + 8 \implies y = -2x + 4$$

2. Case 2: The lines are parallel: This happens when the slopes are identical (when $\frac{-a_1}{b_1} = \frac{-a_2}{b_2} = k$ but y-intercepts differ $\frac{c_1}{b_1} \neq \frac{c_2}{b_2}$

   If we decrease the constant c_1 by 1 unit (so -9 instead of -8), in equation (5), the new first equation in slope/intercept form is: $y = -2x + \frac{9}{2}$.

   This is the equation of a line parallel to the line described in equation (6) $y = -2x + 4$.

   We can plot these using a table (if we wanted to).

3. Case 3: The lines intersect.

   However, if we were to change equation (5) by increasing b_1 by 1 and b_2 by 1, the new equations in slope/intercept form are

$$y = \frac{-4}{3}x + \frac{8}{3} \quad \text{and} \quad y = \frac{-3}{2}x + 3$$

   If we were to solve this system of equations, we would get a solution of $(x, y) = (2, 0)$

   <u>Example:</u> Try another one.

$$\text{(1)} \qquad 2x + 4y - 8 = 0$$
$$\text{(2)}\ x + 2y - 6 = 0$$

1.8 Summation Notation

The Greek letter $\sum$ (sigma) is used to express summation.

Summation notation is used to define the **definite** integral of a continuous function of one variable on a closed interval.

Suppose there is a function $y = x$ but let's say $y = f(x) \implies f(x) = x$. Here f(1)=1, f(2)=2, etc.

Let's now replace x with $i \implies f(i) = i$

If we wanted to sum the values of the function $f(i)$ from $f(1)$ through $f(n)$

$$\sum_{i=1}^{n} f(i) = f(1) + f(2) + f(3) + ... + f(n)$$

$$\sum_{i=1}^{n} f(i) = \sum_{i=1}^{n} i = 1 + 2 + 3 + 4 + 5...n$$

$\sum_{i=1}^{n} f(i)$ is read "Sum of the values of the function $f(i)$ from $f(1)$ through $f(n)$"

Where i is called the index and ranges between 1 and n.

$$\sum_{k=1}^{4}(3k-1) = [3(1)-1] + [3(2)-1] + [3(3)-1] + [3(4)-1] = 26$$

1.8.1 Rules for Summation

Rule 1: The summation of a constant k times a variable is equal to the constant times the summation of that variable.

$$\sum_{i=1}^{n} kX_i = k\sum_{i=1}^{n} X_i$$

Suppose we expand the summation= $kX_1 + kX_2 + kX_3 + ... + kX_n$. Note that we can factor out k and end up with $k(X_1 + X_2 + ... + X_n)$. But $\sum_{i=1}^{n} X_i =$

$(X_1 + X_2 + ... + X_n)$.

Rule 2: The summation of the sum of observations of two variables is equal to the sum of their summations.

$$\sum_{i=1}^{n}(X_i + Y_i) = \sum_{i=1}^{n} X_i + \sum_{i=1}^{n} Y_i$$

If we expand the summations $(X_1 + Y_1) + (X_2 + Y_2) + ...(X_n + Y_n)$. We can rearrange the terms and end up with

$$\sum_{i=1}^{n}(X_i + Y_i) = \sum_{i=1}^{n} X_i + \sum_{i=1}^{n} Y_i$$

Rule 3: The summation of a constant over n observations equals the product of the constant and n

$$\sum_{i=1}^{n} r = rn$$

Let $n = 4 \implies r + r + r + r = 4r$

Another common symbol that you may run across is the product notation symbol, Π. This represents a product of function values. For example

$$\prod_{i=1}^{3} i^2 = (1^2)(2^2)(3^2) = 1 \cdot 4 \cdot 9 = 36$$

1.9 Factorials

The **factorial** function $n!$ is defined for every positive integer n as $n! = n(n-1)\cdots 2 \cdot 1$.

So, for example, $4! = 4 \cdot 3 \cdot 2 \cdot 1 = 24$

The factorial $n!$ gives the number of ways in which n distinct objects can be arranged in order. Each arrangement is called a permutation.

For example, $3! = 6$ since there are six possible permutations of 1,2, and 3: (1,2,3) ; (1,3,2) ; (2,1,3) ; (2,3,1); (3,1,2); and (3,2,1)
It is useful to define $0! = 1$

2 Function Basics

A function is a rule that assigns to each object in a set A exactly one object in a set B.

It is often convenient to represent a functional relationship by an equation $y = f(x)$, which we have done. In this context x and y are called variables. Since the numerical value of y is determined from that of x, we refer to y as the **dependent variable** and x as the **independent variable**.

Suppose you want to find $f(3)$ if $f(x) = 8x - 5 \implies f(3) = 8(3) - 5 = 19$

Often a function is described as a *mapping* from the domain to the range.

Functions assign outputs to inputs. The domain of a function is the set of all possible inputs for the function, say x. The range of a function is the set of its possible output values, say y.

For example: What is the domain and range of $y = f(x) = x^2$? Well, x can take on any value, meaning the domain would be all real numbers. How about the range? For any x that is mapped into y, you get a nonnegative value. Thus, the range would be $y \geq 0$ (all non-negative real numbers).

2.1 Composite Functions

There are many situations in which a quantity is given as a function of one variable that, in turn, can be written as a function of a second variable. By combining the functions in the natural way, you can express the original quantity as a function of the second variable. The result is a **composite function**.

Definition of a composite function: Given two functions f and g, the **composition** of f and g, denoted $f \circ g$ has the value $f(g(x))$. In other words, we evaluate g at x to obtain $g(x)$ and then evaluate f at $g(x)$ to obtain $f(g(x))$.

Note that $f(g(x))$ is only meaningful if x is in the domain of g and $g(x)$ is in the domain of f.
Note that in general, $g(f(x)) \neq f(g(x))$.[2]

Example: Given the functions $f(x) = x^2 + 6$ and $g(x) = 2x - 1$, find $(f \circ g)(x)$.

$$f(g(x)) = (2x-1)^2 + 6 = (2x-1)(2x-1) + 6 = 4x^2 - 4x + 1 + 6 = 4x^2 - 4x + 7$$

Example: Environmental Policy (Homework)

An environmental study of a certain community suggests that the average daily level of carbon monoxide in the air will be $c(p) = 0.5p + 1$ parts per million when the population is p thousand. It is estimated that t years from now the population of the community will be $p(t) = 10 + 0.1t^2$.

a) Express the level of carbon monoxide in the air as a function of time. $\Longrightarrow$ find $c(p(t))$.

$$c(p(t)) = 0.5(10 + 0.1t^2) + 1 = 5 + .05t^2 + 1 = 6 + .05t^2$$

b) When will the carbon monoxide level reach 6.8 parts per million.

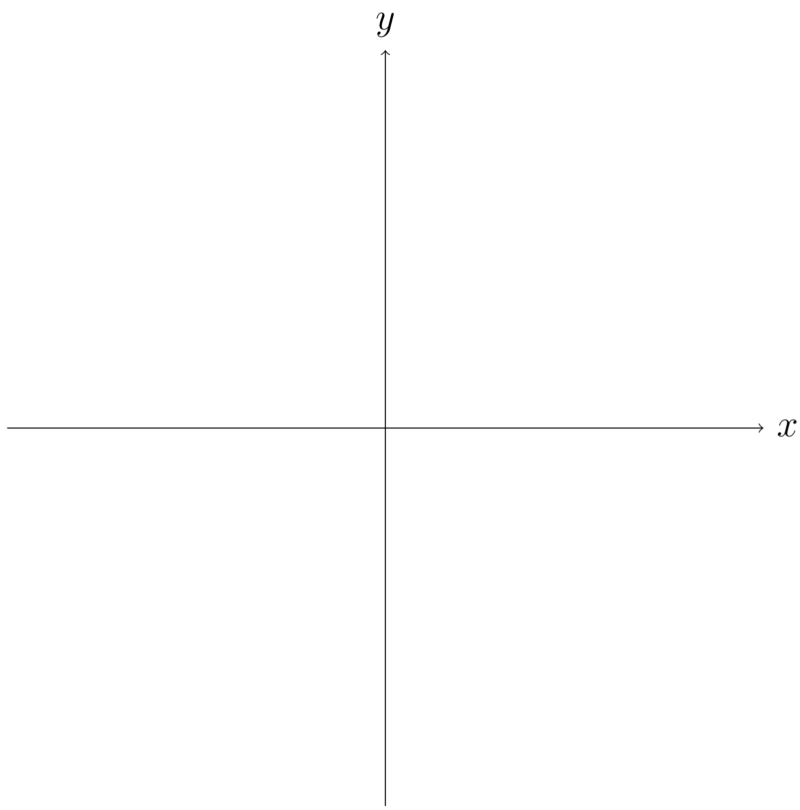
$$c(p(t)) = 6.8 = 6 + .05t^2 \implies .8 = .05t^2 \implies t = (\frac{.8}{.05})^{\frac{1}{2}} = 4$$

2.2 Graphing

The graph of a function f is the spatial or geometric representation of that function. A figure ties together x and $f(x)$ for each x in the domain of f.

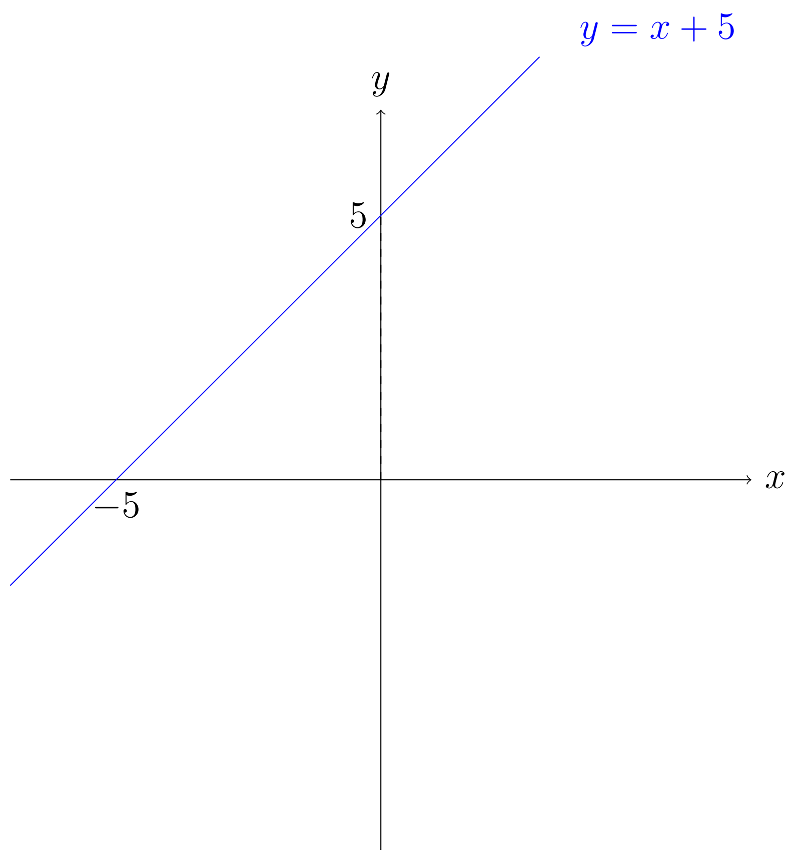
[2]Exception $f(x) = x + 1, \quad g(x) = x - 1$

A rectangular coordinate system



For most functions, we can get a good idea of what they look like by plotting points and charting them on a graph. For example, we can graph the function $f(x) = x+5$ by picking some representative points from the domain, solving their values and plotting them on the coordinate system.
Plot points

x	-3	-2	-1	0
$y = f(x)$	2	3	4	5



The points where a graph crosses the x-axis are called the **x-intercepts** and where the graph crosses the y-axis the **y-intercepts**.

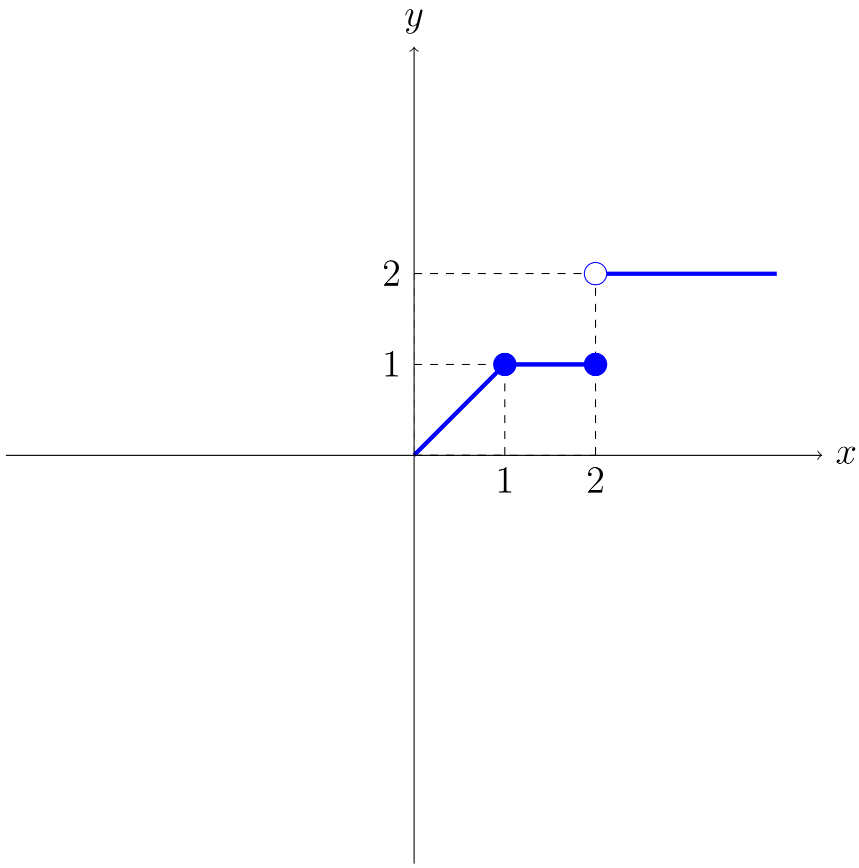
To find the x-intercepts of a function set y equal to zero and solve for x. To find the $y-$intercept of a function set x equal to zero and solve for y.
Find the x and y intercepts

a) y=x+10

b) y=x

We can also plot *stepwise* functions.

$$y = \begin{cases} x & \text{when } 0 \leq x \leq 1 \\ 1 & \text{when } 1 < x \leq 2 \\ 2 & \text{when } 2 < x \end{cases} \tag{7}$$



The open dot indicates that the point is not actually on the graph.

Graphing Non-linear Functions

The graph of $Ax^2 + Bx + C$ is a parabola as long as $A \neq 0$. Parabolas given by the above equation have a "U" (or upside down U) shape.

A parabola opens up if $A > 0$ and opens down if $A < 0$.

The peak or valley of the parabola is called its **vertex**, and it occurs where $x = \frac{-B}{2A}$. To graph a parabola, we need three key pieces of information:

(i) The location of the vertex

(ii) Whether the parabola opens up or down

(iii) Any intercepts

Example: Let's graph $y = x^2 + 3$.

(i) Identify A, B, and C. $A = 1, B = 0, C = 3$.

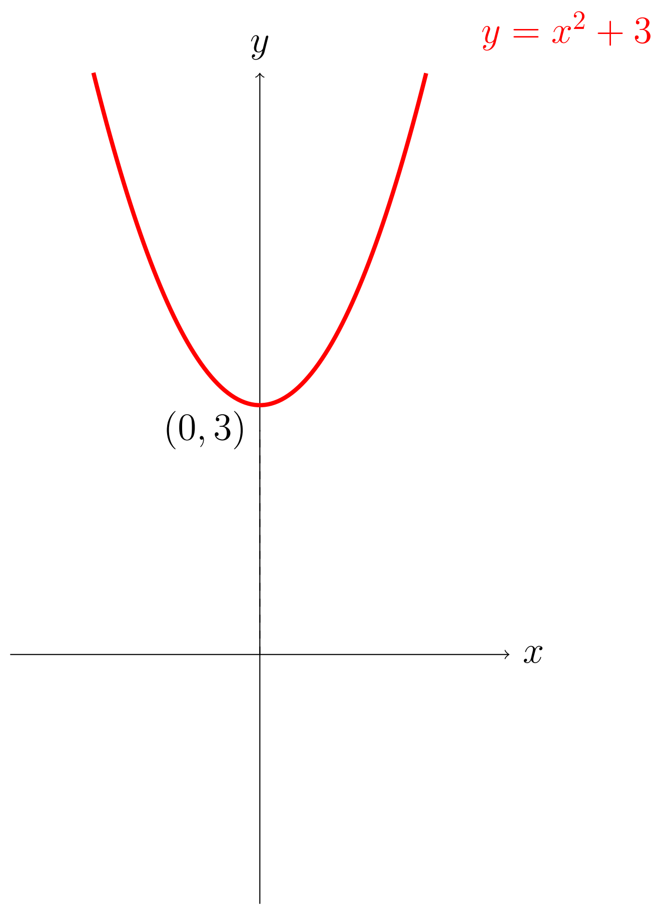
(ii) The location of the vertex: $x = \frac{-B}{2A} = \frac{0}{2} = 0$. Now plug zero into get the y coordinate. $y = 0^2 + 3 = 3$ The vertex is at (0,3).

(iii) Whether the parabola opens up or down: up since $A > 0$

(iv) Any intercepts:

$$x = \frac{0 \pm \sqrt{0 - 4(1)(3)}}{2} = x = \frac{\pm\sqrt{-12}}{2} \implies \text{complex solution.}^3$$

No real roots $\implies$ does not cross x-axis



[3] Or solve directly from $x^2 + 3 = 0 \implies x = \pm\sqrt{-3}$.

Example: Earnings

Suppose the relationship between age (x) and earnings (y) can be expressed as $y = 50x - \frac{1}{2}x^2$.

a) At what age do earnings peak?

Identify $A, B, C \to A = -\frac{1}{2}, B = 50, C = 0$.

Find the vertex $x = \frac{-50}{-1}$ =50. So, earnings peak at age 50. To find y coordinate plug 50 into the function $y = 50(50) - \frac{1}{2}(50)^2 = 1250 \implies$ peak $(x, y) = (50, 1250)$

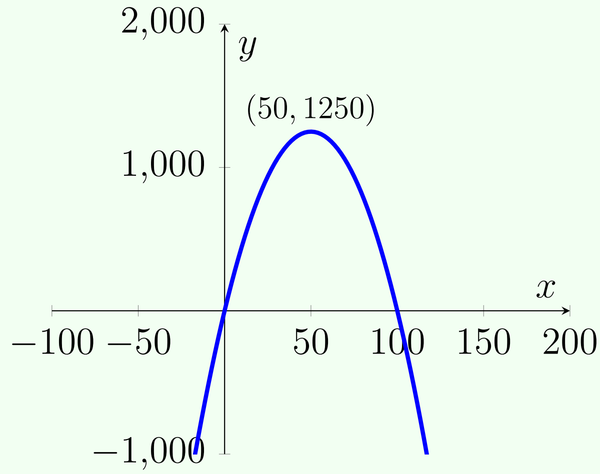
b) Graph the function

Find intercepts

$$x = \frac{-50 \pm \sqrt{50^2 - 4(-.5)(0)}}{-1} = x = \frac{-50 \pm \sqrt{2500}}{-1} = \frac{-50 \pm 50}{-1}.$$

$$x = \frac{-50+50}{-1} = 0 \text{ and } x = \frac{-50-50}{-1} = 100$$

The function crosses the x-axis at 0 and 100. The peak is at (50, 1250) and we know it opens down because $A < 0$



3 Intersection of Functions

The intersection of functions on a graph gives us key information. Specifically, the values of x for which two functions $f(x)$ and $g(x)$ intersect.

To find this algebraically, set $f(x)$ equal to $g(x)$ and solve for x.

Example: Find all intersection points for $f(x) = 5x + 3$ and $g(x) = 3x$

Set $f(x) = g(x) \implies 5x + 3 = 3x \implies 2x = -3 \implies \boxed{x = \dfrac{-3}{2}}$. To find y plug into either function $g(\frac{-3}{2}) = \frac{-9}{2}$

Application to Demand

We can use what we have learned from graphing and finding intersections to the Supply and Demand models you will see in your micro course.

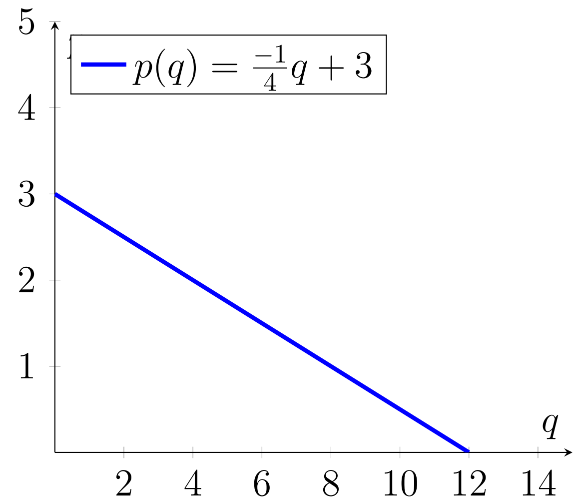
Quantity demanded is the amount of goods buyers are willing and able to purchase.

Law of Demand: $P \uparrow \implies q^D \downarrow$

Consider the following demand schedule for smoothies:

price (p)	quantity (q)
0	12
0.50	10
1	8
1.50	6
2	4
2.50	2
3	0

The function that gives us these values can be expressed as $p(q) = \frac{-1}{4}q + 3$ We can plot these points to get the demand curve.



The table and figure show the relationship between P_m and the quantity demanded of smoothies. **The demand curve slopes down because of the law of demand**.

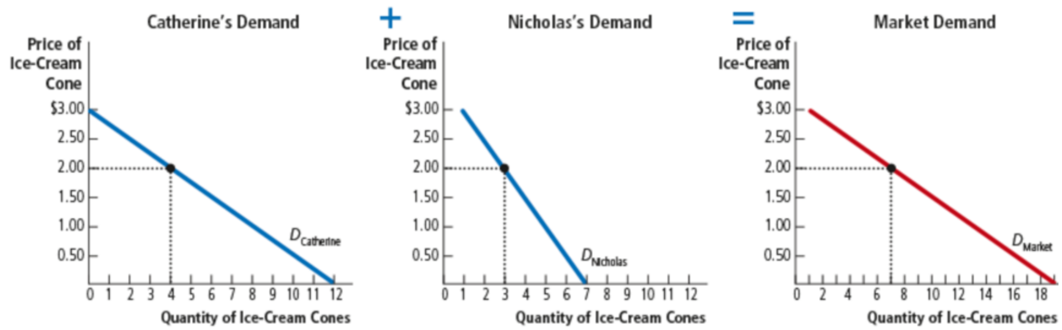
Market Demand

In theory, how do we get to market demand? Well, we can add up everyone's demands.

price (p)	Catherine (q_1)	Nicholas (q_2)	Market
0	12	+ 7	19
0.50	10	6	16
1	8	5	13
1.50	6	4	10
2	4	3	7
2.50	2	2	4
3	0	1	1

Add horizontally

Price of Ice-Cream Cone	Catherine		Nicholas		Market
$0.00	12	+	7	=	19 cones
0.50	10		6		16
1.00	8		5		13
1.50	6		4		10
2.00	4		3		7
2.50	2		2		4
3.00	0		1		1



To analyze markets, we need the market demand, which is the sum of all individual demands.

$$Q_{mkt} = \sum_{\forall i} q_i = q_1 + q_2$$

Since [4]

$$p(q_2) = \frac{-1}{2}q_2 + \frac{7}{2} \implies q_2 = -2p + 7 \text{ and } q_1 = -4p + 12$$

$$\implies \sum_{\forall i} q_i = -4p + 12 - 2p + 7$$
$$Q_{mkt} = -6p + 19$$

We can also do something similar on the supply side, adding all individual supply functions. Suppose we get a market supply of $Q_S = 6p - 5$.

To find the intersection, we would set $Q_S = Q_D$

$$6p - 5 = -6p + 19$$

$$\implies 12p = 24 \implies \boxed{p^* = 2, Q^* = 7}$$

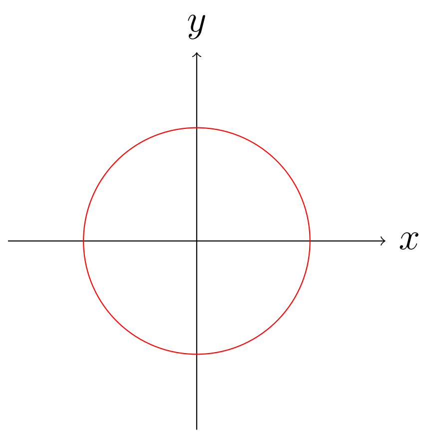
[4]Technically not all the way correct since $q_1 = 0$ for a price above \$3. Similarly, $q_2 = 0$ for $p \geq 3.50$. Here, you have to consider the range of each function.

Vertical line Test

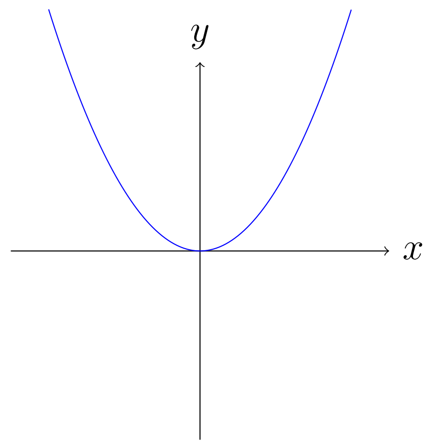
The vertical line test determines whether or not a curve is a graph of a function. Specifically, a curve is the graph of a function if and only if no vertical line intersects the curve more than once. If an x coordinate has more than one y coordinate, then this function fails the vertical line test.

If a graph fails the vertical line test, it is not a function because it does not satisfy the requirement that each x value maps to exactly one y value.[5]

Figure 2: Vertical Line Test Examples



(a) Fails Vertical Line Test (Circle)



(b) Passes Vertical Line Test

If horizontal line test fails, then inverse will fail.

4 Linear Functions

The graph of a linear function is a straight line. Specifically, it is a function of the form: $f(x) = a_0 + a_1x$, where $a_0, a_1 =$ constants.

Linear functions are traditionally written in the form $y = mx + b$, where $m, b =$ constants (**slope-intercept form**).

[5]We call the expression a "relation."

We now show that the parameters m and b are the slope and y-intercept of the line.

The **slope of the line** measures the "steepness" of the line. The formula for computing the slope is

$$Slope = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1}$$

The sign and magnitude of the slope of a line indicate the line's direction and steepness, respectively.

- The slope is positive if the height of the line increases as x increases.
- The absolute value of the slope is large if the slant of the line is severe and small if the slant of the line is gradual.

Example: Find the slope and y-intercept of the line $y + 4x = 6$

$$\implies y = -4x + 6 \implies \text{that the } y-\text{intercept is } \ b = 6 \text{ or } (0, 6). \text{ The slope is } -4.$$

Another way to represent a line is the **point-slope form of the equation of a line.** Specifically, the equation can be expressed as

$$y - y_0 = m(x - x_0)$$

where this is the equation of the line that passes through the point (x_0, y_0) and has a slope equal to m.

Example: Find the equation of the line that passes through the point (6,2) and whose slope is equal to $\frac{1}{5}$.

$$y - 2 = \frac{1}{5}(x - 6) \implies y = \frac{1}{5}x - \frac{6}{5} + 2 \implies y = \frac{1}{5}x + \frac{4}{5}$$

Elasticity and Slope

Continuing with the price and quantity example, we had $P \uparrow$ and $Q_D \downarrow$, but by how much and what can determine how much?

Price Elasticity of Demand: How much buyers respond to changes in market conditions (i.e., magnitude).

- Large change in quantity $\rightarrow$ elastic
- Small change in quantity $\rightarrow$ inelastic

What affects ϵ_P^D? Availability of close substitutes.

How do we compute elasticity? ϵ_P^D= percent change in quantity demanded divided by % Δ in price.

$$\epsilon = \frac{\%\Delta Q}{\%\Delta P}$$

For example suppose that a 10% increase in smoothies $\implies$ quantity $\downarrow$ by 20%

The change in quantity demanded of smoothies is twice as large as the change in price, which is pretty elastic.

- **Point vs Arc Elasticity:**

Example

Consider two points:

$$\text{Point } A\text{: } P = \$2 \text{ and } Q = 120$$
$$\text{Point } B\text{: } P = \$6 \text{ and } Q = 80$$

Compute ϵ_p:

1. $B \to A : \epsilon_p = \frac{120-80}{80} / \frac{2-6}{6} = \frac{1/2}{-(2/3)} = -\frac{3}{4}$
2. $A \to B : \epsilon_p = \frac{80-120}{120} / \frac{6-2}{2} = \frac{-(1/3)}{2} = -\frac{1}{6}$

ϵ_p changes depending on where we start this because the % is calculated from a different base.

To ameliorate this we can use the mid-point method.

$$\text{Let } \bar{Q} = \frac{Q_\theta + Q_\gamma}{2} = 100 \text{ and } \bar{P} = \frac{P_\theta + P_\gamma}{2} = 4$$

$$\therefore \epsilon_p = \frac{120-80}{\bar{Q}} / \frac{2-6}{\bar{P}} = \frac{\frac{40}{100}}{(-1)} = -\frac{40}{100} = -\frac{2}{5}$$

Arc Elasticity: $\frac{\Delta Q}{\Delta P} \cdot \frac{\bar{P}}{\bar{Q}}$
First notice that either way the ϵ_p is negative. This is because of the law of demand–inverse relation between price and quantity.

Demand is considered elastic when $|\epsilon_P^D| > 1$. Here, the quantity responds proportionately more than price.

Demand is considered inelastic if $|\epsilon_P^D| < 1$.

We consider the elasticity unit elastic, $|\epsilon_P^D| = 1$ if the percent change in quantity = percent change in price (ie. 1 to 1 type change).

Given that we are talking about changes in quantity, price, and demand ϵ_P^D is related to the slope of the demand curve.

$|\epsilon_P^D| > 1 \Leftrightarrow$ Elastic $\Leftrightarrow$ Flatter demand curve.

$|\epsilon_P^D| < 1 \Leftrightarrow$ Inelastic $\Leftrightarrow$ Steeper demand curve.

To avoid confusion sometimes we report ϵ_P^D as a positive number. If $\epsilon_P^A = -2$ and $\epsilon_P^B = -1/2$. If I say the larger number is more elastic. Technically ϵ_P^B but we care about magnitude. Therefore we just report absolute value to be on same page.

Elasticity Changes Along Demand

As you move down the demand curve ϵ_p becomes more inelastic.

Intuition: At high prices a slight percent increase in price $\implies$ stop buying. At lower prices the same percent increase is not breaking the bank.

Steeper slope $\implies$ the less elastic the demand.

Slope and Elasticity:

$$\epsilon_p^D = \frac{\%\Delta Q}{\%\Delta P} \implies \frac{\Delta Q}{Q} / \frac{\Delta P}{P} = \frac{\Delta Q}{Q} \times \frac{P}{\Delta P} = \frac{\Delta Q}{\Delta P} \times \frac{P}{Q} = \underbrace{\frac{\Delta Q}{\Delta P}}_{\text{Inverse of Slope}} \times \frac{P}{Q}$$

Recall slope $\frac{\Delta Y}{\Delta X} \implies \frac{\Delta P}{\Delta Q} \implies$ Inverse of $\frac{\Delta Q}{\Delta P}$

$\implies$ slope $\neq \epsilon_P^D$. $\frac{\Delta Q}{\Delta P}$ is constant, but ϵ_P^D changes as you move along the demand, depends on $\frac{P}{Q}$.

A 10% increase at \$2 is \$2.2 $\implies$ 20 cents

A 10% increase at \$1 million is \$100,000 $\implies$ \$1.1 million

Parallel and Perpendicular Lines

Specifically, let m_1 and m_2 be the slopes of the nonvertical lines L_1 and L_2. Then,

L_1 and L_2 are **parallel** iff $m_1 = m_2$
L_1 and L_2 are **perpendicular** iff $m_2 = -\frac{1}{m_1}$

Example: Let L be the line $x + y = 2$

a) Find the equation of a line parallel to L through point (1,2). The slope of L is -1. Use point-slope formula.

$$y - 2 = (-1)(x - 1) \implies y = -x + 1 + 2 \implies y = -x + 3$$

b) Find the equation of a line perpendicular to L through point (-5,7). Perpendicular slope is +1.

$$y - 7 = (1)(x - (-5)) \implies y = x + 5 + 7 \implies y = x + 12$$

Distance Formula

The **distance formula** tells us the distance between two points (x_1, y_1) and (x_2, y_2). The formula is

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Example: Find the distance between the points (2,1) and (-5,3)

$$\text{Distance} = \sqrt{((-5) - 2)^2 + (3 - 1)^2} = \sqrt{49 + 4} = \sqrt{53}$$

5 Exponential and Logarithmic Functions

5.1 Exponential Functions

In previous lessons, we dealt with power functions such as $y = x^2$, in which the variable x is raised to a constant power 2 (the exponent).

Now we'll be looking at functions for which a constant base like 2 is raised to a variable x so $y = 2^x$. These are called **exponential functions.** If $b > 0$, the function, called the exponential function with base b, is defined by

$$f(x) = b^x \quad \text{for every real number } x$$

We note that $b = 1$ is not an interesting case and we won't deal with it. Exponential functions are common across disciplines.

Here is a refresher of power laws of b^n for rational values of n (and $b > 0$)

- Integer Powers: If n is a positive integer, then $b^n = b \cdot b \cdot b \cdot \cdots \cdot b$, where this product contains n factors.
- Fractional Powers: If n and m are positive integers, then $b^{\frac{n}{m}} = (\sqrt[m]{b^n}) = (\sqrt[m]{b})^n$
- Negative powers: $b^{-x} = \frac{1}{b^x}$
- Zero power: $b^0 = 1$

Now we can state the basic properties of exponential functions:

For bases, a, b $(a > 0, b > 0)$ and any real numbers x, y we have

- The **equality rule**: $b^x = b^y$ iff $x = y$
- The **product rule**: $b^x b^y = b^{x+y}$
- The **quotient rule**: $\frac{b^x}{b^y} = b^{x-y}$
- The **power rule**: $(b^x)^y = b^{xy}$
- The **multiplication rule**: $(ab)^x = a^x b^x$
- The **division rule**: $\left(\frac{a}{b}\right)^x = \frac{a^x}{b^x}$

Example: Simplify the following

a) $s^2 s^{-1} = s$

b) $\frac{b^{2x}}{b^{4x}}$ $=b^{2x-4x} = b^{-2x}$

c) $(8^{\frac{1}{2}})(6^{\frac{1}{2}}) = (8 \cdot 6)^{\frac{1}{2}} = 48^{\frac{1}{2}}$

In calculus, the most common exponential base is denoted by the letter e. This letter is one of the most important in all of mathematics.[6] e is an irrational number that roughly equals 2.72. It is frequently referred to as the "natural exponential base." We can use limits (discussed later) to formally express it but we can also use the tools we have learned thus far

$$e = \sum_{n=0}^{\infty} \frac{1}{n!}$$

5.2 Logarithmic Functions

If N is a positive number and b is also positive (with $b \neq 1$), we may ask if there is a number L such that $b^L = N$. It can be shown that there is exactly one such number L.

As L depends on both b and N, the notation developed for it is $L = \log_b N$. L is called **the logarithm of** N **to the base** b.

L is the exponent that b must be raised to yield N. If instead of constant N, we let the number of N vary (call it x instead of N).

The **logarithm** of x to the base b is y such that $b^y = x$ and we write $y = \log_b x$ (where $b > 0$ and $b \neq 1$). In the special case where $b = e$, the number y is such that $e^y = x$ is called the natural logarithm $y = \ln x$.

$$\boxed{y = \ln x \;\text{ iff }\; e^y = x}$$

Example: Change the following exponential forms to logarithms

[6] The name "e" for the constant was coined by Euler, who studied the properties of the exponential function e^x in the mid-18th century and found it to be a fundamental mathematical constant. However, it was first introduced by the Scottish mathematician John Napier in the early 17th century

a) $81 = 9^2 \implies \log_9 81 = 2$

b) $\frac{1}{9} = 3^{-2} \implies \log_3 \frac{1}{9} = -2$

<u>Example:</u> Solve for the unknown quantity in simplest terms

a) $y = \log_{30} 900$. Here, y is the exponent and b, 30, must be raised to get N, 900. Thus $30^y = 900 \implies y = 2$

b) $y = \log_2 \frac{1}{32}$. Here, y is the exponent and b, 2, must be raised to get N, $\frac{1}{32}$.

$$2^y = \frac{1}{32} \implies \frac{1}{2^{-y}} = \frac{1}{32}.$$

How many times do you have to raise 2 to in order to get 32 in the denominator? 5 $\implies y = -5$

The natural logarithm ($\ln x$) and the expression e^x have a "neutralizing" effect (cancel each other out; note that $e^{\ln x} = x$ requires $x > 0$).

$$\ln e^x = e^{\ln x} = x$$

The natural log of e raised to x is simply x and e raised to the power $\ln x$ is simply x.

This simply tells us that the consecutive impact on x of the exponential function followed by the natural log function leaves x unchanged.

Simple Proof
Claim is that $\ln e^x = x$.

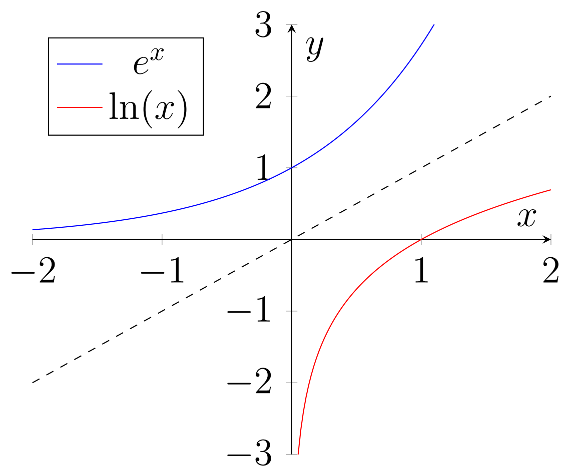
Let $\log_e e^x = u$. Recall that in this case $b = e$, $L = u$, and $N = e^x$.

Remember that we interpret L (u) as what $b(e)$ needs to be raised to in order to get N (e^x). In other words, $b^L = N$ or in this case $e^u = e^x$. Now, recall the equality rule $s^x = s^y$ iff $x = y$. This $u = x \implies \log_e e^x = x \implies \ln e^x = x$

5.3 Inverse Functions

This "cancellations" between e^x and $\ln x$ happens because the functions are actually inverses of each other.

In general, two functions f and g for which $f(g(x)) = x$ and $g(f(x)) = x$ are said to be **inverses** of each other. Thus the exponential function and the logarithmic function are inverses.



Notice that the graph of $y = \ln x$ is obtained by reflecting the graph of $y = e^x$ across the line $y = x$.

Properties of logarithms

- The **equality rule:** $log\ u = logv$ iff $u = v$
- The **product rule**: $log\ uv = log(u) + log(v)$
- The **power rule**: $log\ u^r = r \cdot logu$
- The **quotient rule**: $log\frac{u}{v} = log\ u - log\ v$

Note these rules apply to all logs not just those based e.

Simple Proof: Let's prove the product rule so that we can get the intuition behind how these rules work:

Let $m = \log_b\ u$ and $n = \log_b\ v$, so $u = b^m$ and $v = b^n$

$$\implies \log_b\ uv = \ \log_b b^m b^n \ = \ \log_b b^{m+n} \ = \ m + n = \log_b\ u + \log_b\ v$$

<u>Examples</u> Solve each for x in terms of y

a) $\log_a x = y^3$

$$\implies a^{\log_a x} = a^{y^3} \implies x = a^{y^3}$$

b) $\log_a x = \log_a 3 + \log_a y$

$$\implies \log_a x = \log_a 3y \implies a^{\log_a x} = a^{\log_a 3y} \implies x = 3y$$

c) $\ln x = \log_a y$

d) $y = ge^{hx}$

<u>More Examples</u>

Simplify each of the following

a) $e^{3\ln x}$

b) $e^{\frac{1}{2}\ln 6x}$

<u>More Examples</u>

Use the properties of logs to write the following as sums, differences, or products.

a) $\ln 76x^3$

b) $\ln x^5y^2$

c) $\ln 12 + \ln 5 - \ln 6$

d) $5\ln \frac{1}{2}$

There also exists a **conversion (or base change) formula** that converts logs of one base to logs of another base.

Specifically, if a and b are bases and $b \neq 1$ then

$$\log_b x = \frac{\log_a x}{\log_a b}$$

<u>An Example</u>

Using the logarithmic conversion formula, we want to find $\log_{10}(100)$, which means we want to find the logarithm of 100 with base 10.

The logarithmic conversion formula for base change is:

$$\log_a(b) = \frac{\log_c(b)}{\log_c(a)}$$

In this case, we have $a = 10$ and $b = 100$. To find $\log_{10}(100)$, we can use any base for c, but a common choice is $c = e$. So, we can rewrite the equation as follows:

$$\log_{10}(100) = \frac{\log_e(100)}{\log_e(10)}$$

Now, let's evaluate the logarithms on the right-hand side:
1. $\log_e(100) \approx 4.605$ 2. $\log_e(10) \approx 2.302$
Finally, we can calculate the value of $\log_{10}(100)$ using the base change formula:

$$\log_{10}(100) = \frac{\log_e(100)}{\log_e(10)} \approx \frac{4.605}{2.302} \approx 2$$

So, $\log_{10}(100) \approx 2$.